

Road Accident Data Analysis and Hotspot Identification System

Accident Overview

Overview

Risk Analysis

Hotspot Analysis

12K

Total Accidents

158

Fatal Accidents

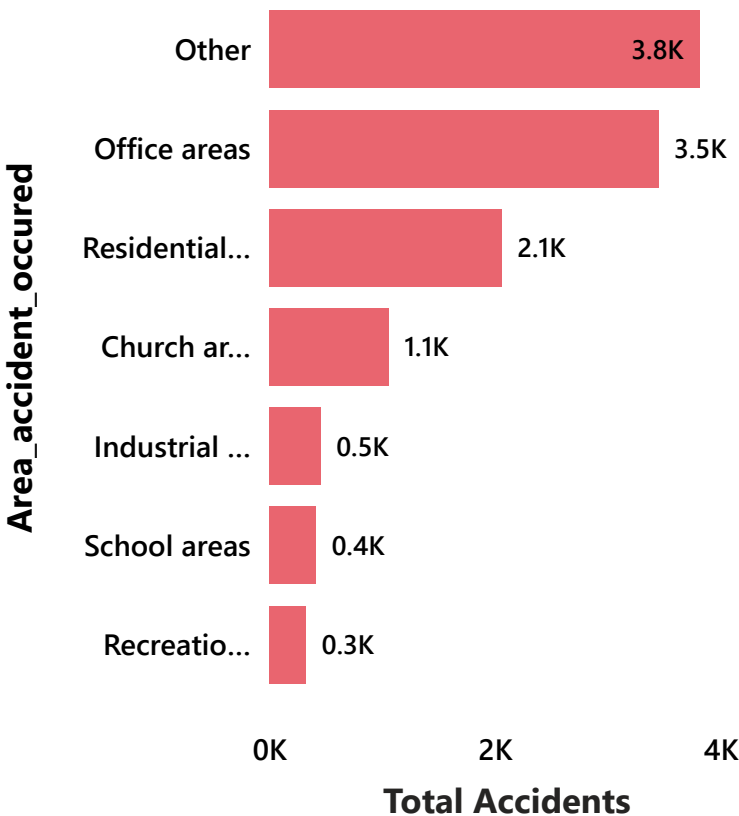
1.55

Avg Casualties

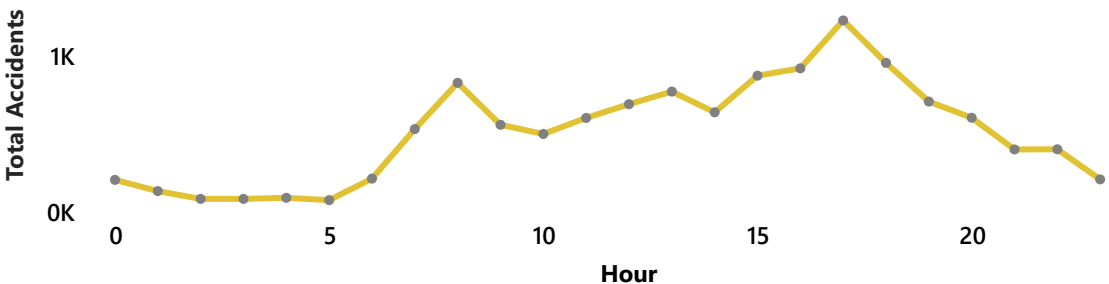
17

Peak Hour

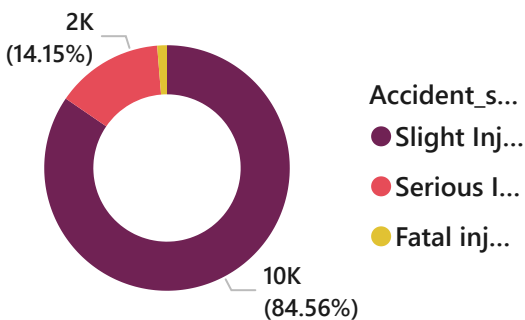
Total Accidents by Area Type



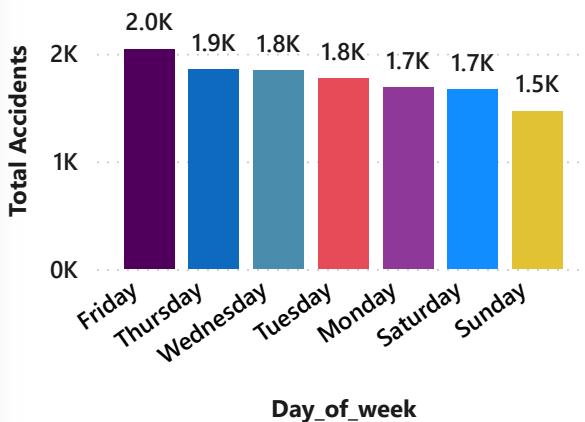
Total Accidents by Hour



Accident Severity Distribution



Total Accidents by Day of the Week



Approximately 12,000 accidents have been recorded. Most accidents occur during peak evening hours, especially around 5 PM. Urban and office areas report the highest number of accidents due to heavy traffic conditions. While the majority of accidents are of slight severity, serious and fatal cases are also observed.

Risk analysis

[Overview](#)[Risk Analysis](#)[Hotspot Analysis](#)**11**

Weather Risk

29

Night Accident %

20

Unique Causes

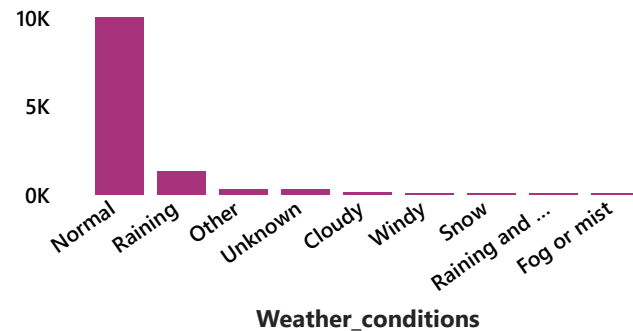
24

Road Risk %

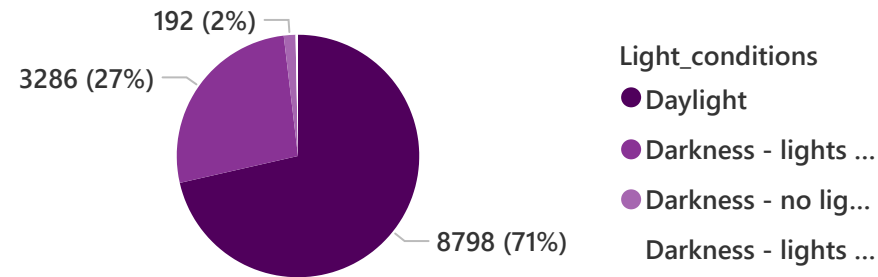
12,316

Vehicle Count

Weather Impact on Accidents



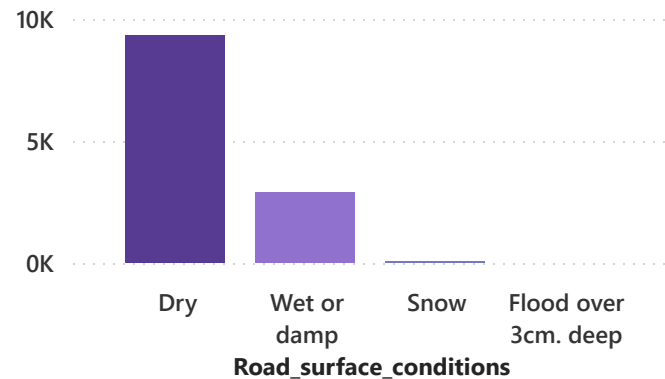
Accidents by Light Conditions



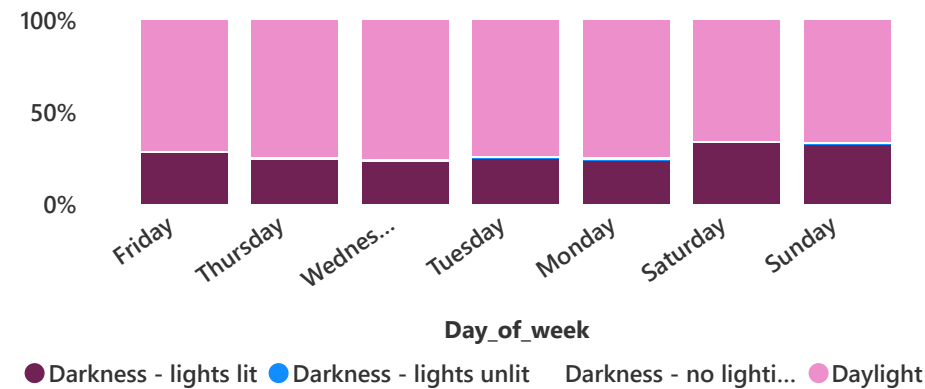
Total Accidents by Cause_of_accident



Road Surface Risk Analysis



Total Accidents by Day_of_week and Light_conditions



Accidents mostly occur during normal and rainy weather conditions, indicating that driver behavior plays a significant role. A large proportion of accidents happen during daylight due to increased traffic volume. Poor driving practices such as lack of safe distance, careless driving, and improper lane changes are identified as the major causes. Road surface conditions like wet or damp roads also contribute to accident risk.

Hotspot Analysis

Overview

Risk Analysis

Hotspot Analysis

Weather_conditions

All

14375

Severity Score

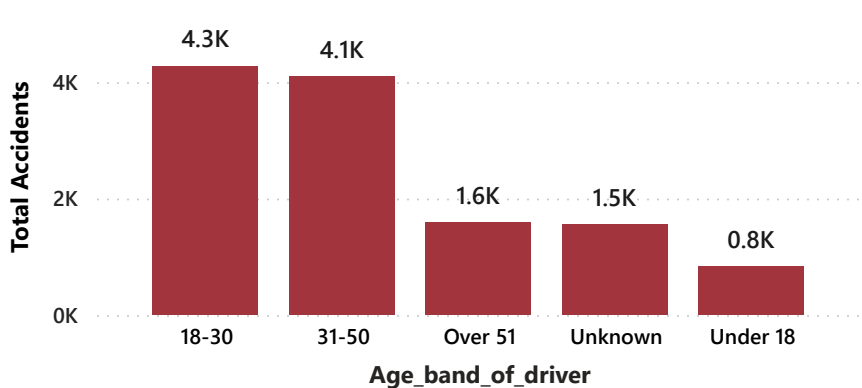
100

High Risk %

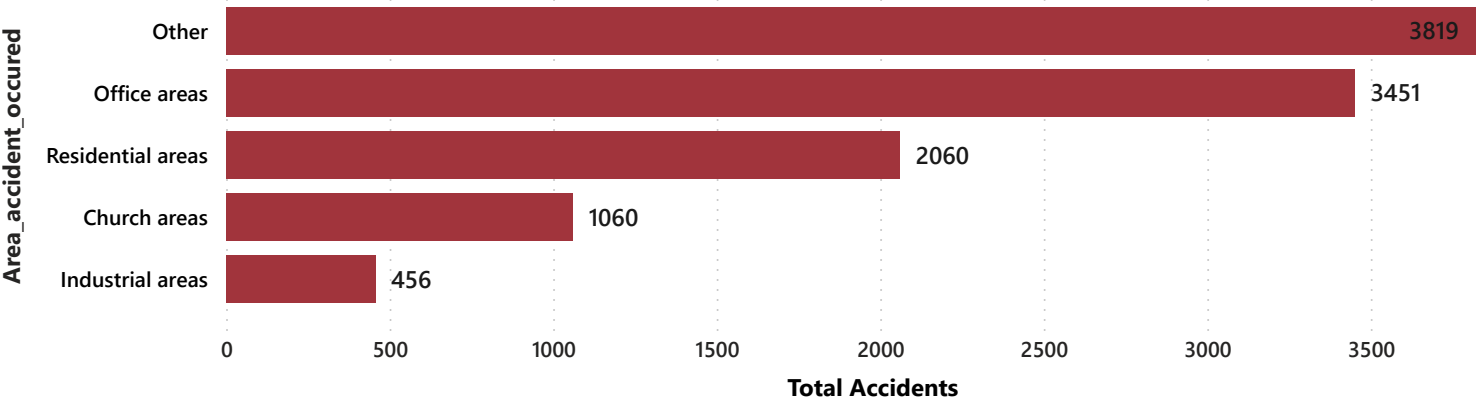
Other

Predicted Hotspot

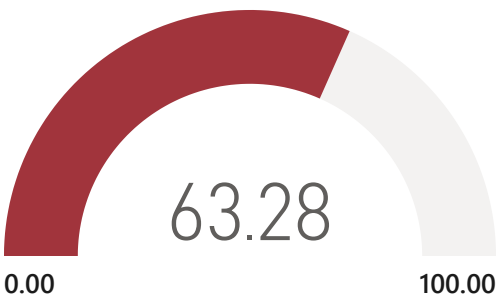
Accidents by Driver Age Group



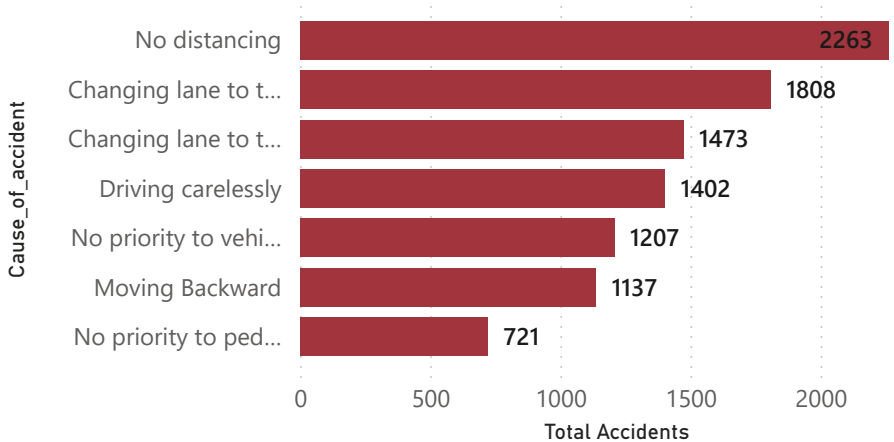
Accident Hotspots by Area Type



Risk Score Percentage



Accidents by Cause



The analysis identifies urban areas as high-risk zones with the highest number of accidents. Drivers aged between 18 and 50 are more frequently involved in accidents. The main contributing factors include careless driving, failure to maintain safe distance, and improper lane usage. Overall, the risk score indicates a moderate to high level of accident risk in these regions.